Explicit String Vacua and the Geometric Selection Conjecture

By Deep Bhattacharjee and Ushashi Bhattacharya

Abstract

We study what an explicit construction would have to establish for the Geometric Selection Conjecture. We give a smooth globally proper selection functional once a compatible viable point is supplied, distinguish this marked-point construction from a selector intrinsic to an unmarked metric, and prove a quantitative criterion for simultaneous stabilization and preservation of physical mass inequalities. We independently check the slope, Chern-class and index data of a published tetraquadric line-bundle example. An explicit normalization example shows that a holomorphic Yukawa hierarchy need not be a physical hierarchy. A local supersymmetric effective theory has an isolated minimum with broken supersymmetry, but its realization by the compactification is not established. No vacuum satisfying all nine conditions is constructed, and the full conjecture remains open.

Research status. This is an author-review draft for a possible revision of [3]. It is not a proof of the full Geometric Selection Conjecture (GSC). The use of an Annals manuscript class concerns formatting; no journal acceptance, submission, or new Hexagon identifier is asserted. The elementary construction criteria below are not claimed to be previously unknown theorems.

1. The simultaneous construction problem

The GSC formulation in [2] and [3] concerns a landscape $\mathcal{L} = \coprod_\tau M_\tau$ of compactification data and an involutive isometry μ defined on a union D of components. A desired function $S : \mathcal{L} \to [0, \infty)$ is smooth and proper on each component, has a unique global minimum at a viable point, and is invariant under μ wherever μ is defined. Its viability requirements originate in [1]. We

Received by the editors 25 October 2026. Author-review draft; the full GSC remains unproved.

2020 *Mathematics Subject Classification.* Primary 14J32, 81T30; Secondary 53C25.

Keywords: heterotic compactification, Geometric Selection Conjecture, line bundles, moduli stabilization, physical Yukawa couplings

use the following short restatement, without supplying new numerical meanings for the physical acceptance conditions.

C1	A slope-polystable bundle with $c_1 = 0$ and an effective anomaly class.
C2	Three net families, with the quotient index treated consistently.
C3	Wilson-line reduction of the visible gauge group to the Standard Model.
C4	Three quark–lepton families and no chiral exotics after projection.
C5	Allowed sufficiently heavy masses for vectorlike exotics.
C6	A Higgs pair with doublet–triplet splitting.
C7	Stabilization of the relevant moduli in a controlled effective regime.
C8	Acceptable physical Yukawa rank and hierarchy after normalization.
C9	Phenomenologically consistent supersymmetry breaking, if present.

These are nine simultaneous predicates on one complete datum, not nine alternative cases. Let A_i be the locus satisfying Ci. The existence problem is $\bigcap_{i=1}^{9} A_i \neq \varnothing$. Nonemptiness of separate A_i, even of all pairs among three loci, need not imply a common point: $\{0,1\}$, $\{1,2\}$ and $\{0,2\}$ have pairwise intersections but empty triple intersection. Results at different bundles or different moduli values cannot be assembled into a single vacuum without proving their compatibility.

For a complete datum one must specify the threefold, visible and hidden bundles, polarization, free action and equivariant structures when used, Wilson line, effective Kähler potential and superpotential, and the domain in which omitted corrections are controlled. C5 and C7–C9 also need explicit acceptance predicates. We leave those predicates as inputs; choosing convenient thresholds would change the problem.

2. **A marked-point selection construction**

The functional conditions have a clean differential-topological answer. They do not manufacture the point that is to be selected.

THEOREM 2.1 (Selection with a supplied point). *Let M be a countable disjoint union of second-countable smooth Riemannian manifolds. Let μ be an involutive isometry on a union D of components. Given $p \in M$, the following are equivalent*:

(*i*) $p \notin D$, *or* $\mu p = p$.

(*ii*) *There is a nonnegative smooth globally proper function S on M with unique zero p and $S \circ \mu = S$ on D.*

The function can be chosen to have Hessian $2g_p$ at p.

Proof. An invariant function with unique zero must fix that zero under every symmetry defined there. This proves necessity.

Every second-countable smooth manifold has a nonnegative smooth proper exhaustion ρ. One direct construction takes a compact exhaustion $K_j \subset \operatorname{int} K_{j+1}$, a locally finite partition of unity subordinate to the compact annular cover, and assigns successively increasing positive coefficients to its terms; outside K_j all nonzero coefficients can be arranged to be at least j. Its sublevel sets are closed and contained in some K_j, hence compact. These standard smooth-manifold constructions apply separately to the components; see [7].

On a component preserved by μ, replace ρ by $\rho + \rho \circ \mu$. On a pair of exchanged components choose ρ on one and transport it to the other. The resulting function is invariant and proper on each component. Enumerate the component orbits; outside the orbit of the component of p, add distinct positive integer offsets tending to infinity. Each orbit contains at most two components.

Choose a normal ball $B_r(p)$ with compact closure, and write $f(x) = d(p,x)^2$ there. If $p \in D$ and is fixed, this ball and f are invariant. Choose a smooth function $\theta : [0,\infty) \to [0,1]$ equal to one for $t \le r^2/4$ and zero for $t \ge r^2/2$. On the component of p set

$$S = \theta(f) f + (1 - \theta(f))(1 + \rho),$$

extending the expression by $1 + \rho$ outside the ball. This is smooth because $\theta(f)$ vanishes near its boundary. Its only zero is p: away from p every nonzero term in this convex combination is positive. It equals f near p, so its Hessian there is $2g_p$.

On the remaining components use their exhaustion plus the positive orbit offset. For a fixed sublevel only finitely many offsets occur. On the distinguished component a sublevel is a closed subset of $\overline{B}_r(p) \cup \{\rho \le t - 1\}$; on all other relevant components it is compact by properness of ρ. Thus every sublevel of S is compact in the disjoint union. Invariance follows from the construction. □

Remark 2.2 (What the point supplies). The construction uses the marked point p, exhaustions and cutoffs in addition to the metric. It proves the smooth-function assertion with a supplied compatible point. It does not prove that an unmarked metric determines p or that p satisfies C1–C9.

Proposition 2.3 (An intrinsic selector must respect all isometries). *Suppose a rule assigns S_g to an unmarked Riemannian manifold naturally under isometries. If S_g has a unique minimum p, every isometry fixes p. In particular there is no such uniquely minimizing rule on a positive-dimensional homogeneous Riemannian manifold.*

Proof. Naturality for an isometry h says $S_g(hx) = S_g(x)$. Hence hp is another minimum and equals p. Transitivity on a manifold with more than one point is incompatible with a common fixed point. □

This is an obstruction to a particular notion of metric-only selection, not a proof that the actual heterotic moduli metric is homogeneous. It also shows why invariance under an additional compact group cannot be discarded merely because the partial mirror map is undefined at p: the additional group must still fix a unique selected point.

3. **Index constraints and the mirror restriction**

PROPOSITION 3.1 (Index and finite descent). *Let X be a Calabi–Yau threefold and V a holomorphic bundle with $c_1(V) = 0$. Then*

$$\chi(X, V) = \frac{1}{2} \int_X c_3(V).$$

If $\pi : X \to Y$ is finite étale *of degree d and $V = \pi^* V_Y$, then $\chi(X, V) = d\chi(Y, V_Y)$. Thus the three-family index condition downstairs requires $|\int_X c_3(V)| = 6d$.*

Proof. In Hirzebruch–Riemann–Roch, the degree-six term of $\mathrm{ch}(V)\,\mathrm{td}(X)$ is $\mathrm{ch}_3(V) + c_1(V)c_2(X)/12 + operatornamerk(V)c_1(X)c_2(X)/24$. The last two terms vanish and $\mathrm{ch}_3(V) = c_3(V)/2$. For an *étale* map the tangent bundle and Todd class pull back, and integration of a pulled-back top class multiplies by d. □

For a quintic, expansion of $(1 + H)^5/(1 + 5H)$ gives $c_3(T_X) = -40H^3$ and $\int_X H^3 = 5$, hence $\int c_3(T_X) = -200$. Consequently no standard-embedding free quotient has index of absolute value three, because $100 = 3d$ has no integer solution. This restriction is independent of the quotient group classification. It does not apply to arbitrary nonstandard bundles.

Under the standard-type partial mirror convention of [3], a fixed point has $h^{1,1} = h^{2,1}$ and hence $\chi(X) = 0$. The bundle on that component has $\int c_3(V) = \chi(X)$, so Proposition 3.1 gives zero net families. This proves that a selected viable point cannot be a mirror-fixed point in that specified domain. It does not exclude nonstandard data outside the domain, nor establish a mirror map on all $(0, 2)$ compactifications.

4. **An exact nonstandard geometric starting point**

We now check a published bundle example rather than assume that a topological index determines its masses. The bundle data are those of Buchbinder, Constantin and Lukas [4, Sec. 5.1]; the calculations below are independent reproductions, not a new discovery of their model.

Let $X \subset A = (\mathbb{P}^1)^4$ be a smooth hypersurface of multidegree $(2,2,2,2)$. Write J_i for the four hyperplane classes. Adjunction gives $K_X \cong \mathcal{O}_X$ and

$$d_{ijk} = \int_X J_i J_j J_k = \begin{cases} 2 & i,j,k \text{ distinct}, \\ 0 & \text{otherwise}, \end{cases} \qquad c_2(T_X) = 4 \sum_{i<j} J_i J_j.$$

These formulas follow in the ambient ring $J_i^2 = 0$ from $[X] = 2\sum J_i$ and $c(T_X) = \prod_i (1+2J_i)/(1+2\sum J_i)$. Set $V = \bigoplus_{a=1}^{5} \mathcal{O}_X(k_a)$, with columns

$$(k_a^i) = \begin{pmatrix} -1 & -1 & 0 & 1 & 1 \\ 0 & -3 & 1 & 1 & 1 \\ 0 & 2 & -1 & -1 & 0 \\ 1 & 2 & 0 & -1 & -2 \end{pmatrix}. \tag{4.1}$$

Proposition 4.1 (Cover-space data). *For every smooth X above, $c_1(V) = 0$ and V is slope-polystable for $J = t(J_1+J_2+J_3+J_4)$, $t>0$. Its Chern and index data are*

$$\left(\int_X c_2(V) J_i \right)_{i=1}^{4} = (24, 8, 20, 12), \qquad \int_X c_3(V) = -24, \qquad \chi(X,V) = -12.$$

The anomaly class on the cover has the effective representative $6J_1J_3 + 2J_1J_4$.

Proof. Every row of (4.1) sums to zero, proving $c_1(V) = 0$. Every column also sums to zero. The slope of a summand is therefore $12t^2 \sum_i k_a^i = 0$. Rank-one bundles are slope-stable, so their direct sum at common slope zero is polystable.

Put $l_a = \sum_i k_a^i J_i$. From $\sum l_a = 0$ one obtains $c_2(V) = -\frac{1}{2}\sum l_a^2$ and $c_3(V) = \frac{1}{3}\sum l_a^3$. Direct multiplication using d_{ijk} gives the displayed values. Independently, for a line bundle,

$$\chi(\mathcal{O}_X(k)) = 2 \sum_{i<j<k'} k_i k_j k_{k'} + 2 \sum_i k_i.$$

For the five columns the indices are $0, -8, 0, 0, -4$, giving -12. The anomaly pairs with the J_i as $(0, 16, 4, 12)$, exactly the pairings of $6J_1J_3 + 2J_1J_4$. The J_i form an integral basis of $H^2(X)$ by Lefschetz; $H^4(X)$ is torsion-free by Lefschetz on H_2 and Poincaré duality. Equality of these pairings therefore identifies the classes. Each J_iJ_j with $i \neq j$ is represented by an effective intersection of divisors, proving the assertion. □

4.1. *A free degree-four family and equivariance.* On every factor let $a[x:y] = [x:-y]$ and $b[x:y] = [y:x]$ and act diagonally on A. They commute projectively and generate $\Gamma \cong (\mathbb{Z}/2)^2$. Each nonidentity element fixes $2^4 = 16$ ambient points. The three sets are disjoint, so there are 48 points to avoid. Put

$$e_i = x_i^2 + y_i^2, \quad d_i = x_i^2 - y_i^2, \quad p_i = 2x_iy_i.$$

An invariant multidegree-$(2,2,2,2)$ linear system is spanned by products of one of e_i, d_i, p_i per factor with even numbers of d and p.

LEMMA 4.2 (Base-point-free invariant system). *This invariant linear system is base-point-free. Its general member is smooth and avoids all* 48 *fixed points, so* Γ *acts freely on it.*

Proof. At a point of any factor at least two of e, d, p are nonzero. Label them by $0, u, v \in (\mathbb{Z}/2)^2$. Any two-element subset of these three labels is an affine line. If the available two-element sets in four factors have different directions, their sumset is the whole group. If all have the same direction, they are the same pair and their fourfold sumset contains zero. A choice of nonzero factors with total label zero therefore exists, proving base-point-freeness. Taking subsets deals also with a factor where all three values are nonzero.

Bertini gives smoothness on a nonempty open subset of the parameter space. At any of the finitely many fixed points base-point-freeness makes the vanishing condition a proper hyperplane. The complement of these hyperplanes meets that open subset. Such a member has a free action. The product of the four determinants of either ambient lift is $(-1)^4 = 1$, and its defining polynomial is invariant, so the residue holomorphic three-form is preserved. Thus the free quotient has trivial canonical bundle as well. □

The two linear lifts anticommute on $\mathcal{O}(1)$ of one factor. Their commutator on $\mathcal{O}(k_1, k_2, k_3, k_4)$ is $(-1)^{\sum k_i}$. Every column of (4.1) has sum zero, so these tensor lifts commute and square to one. They give equivariant structures on the summands and a descended bundle V_Y on $Y = X/\Gamma$. Proposition 3.1 now proves $\chi(Y, V_Y) = -3$.

The invariant polarization descends and proves polystability downstairs. The anomaly class is rationally effective downstairs: pushing forward an effective representative upstairs and dividing by four proves this over $\mathbb{Q}$. An equality after pullback alone does not exclude integral torsion in $H^4(Y, \mathbb{Z})$. We have not audited that torsion refinement or the hidden-sector completion here. We therefore distinguish the verified rational anomaly statement from a full quantized anomaly certificate for the quotient.

A Wilson homomorphism on the $SU(5)$ factor may send a to $\mathrm{diag}(1, 1, 1, -1, -1)$ and b to the identity. Its centralizer is $S(U(3) \times U(2))$, the Standard Model group up to a finite quotient. This group calculation is not a calculation of all projected cohomology groups, possible additional gauge factors, or their masses. Those are separate C3–C6 obligations. The example is a geometric starting point, not a certificate of a nine-condition vacuum.

5. **Why holomorphic masses do not close C8**

After fixing conventions and the Higgs factor, a physical Yukawa matrix has the form

$$Y_{\mathrm{phys}} = c\, Z_L^{-T/2} Y_{\mathrm{hol}} Z_R^{-1/2}, \qquad c > 0, \tag{5.1}$$

where the matter metrics are positive definite. Their invertibility preserves rank but does not preserve singular-value ratios.

PROPOSITION 5.1 (A normalization counterexample). *For* $0 < \epsilon < 1$, *take*

$$Y_{\mathrm{hol}} = \mathrm{diag}(\epsilon^8, \epsilon^4, 1), \quad Z_L = I, \quad Z_R = \mathrm{diag}(\epsilon^{16}, \epsilon^8, 1), \quad c = 1.$$

All metrics are positive definite and $Y_{\mathrm{phys}} = I$. *Thus a full-rank holomorphic matrix with separated scales does not prove a physical hierarchy.*

Proof. The positive inverse square root of Z_R is $\mathrm{diag}(\epsilon^{-8}, \epsilon^{-4}, 1)$. Multiplication in (5.1) gives the identity. □

The example does not claim that these metrics occur in the tetraquadric model. It refutes the inference based on a holomorphic matrix alone. It does not refute a hierarchy proved with regular nondegenerate metrics at the limiting point, as the following bound makes precise.

PROPOSITION 5.2 (Bounded normalization preserves orders). *Suppose* $mI \preceq Z_L, Z_R \preceq MI$ *with* $0 < m \le M$ *and* $0 < c_- \le c \le c_+$. *For every singular value, ordered consistently,*

$$\frac{c_-}{M}\sigma_i(Y_{\mathrm{hol}}) \le \sigma_i(Y_{\mathrm{phys}}) \le \frac{c_+}{m}\sigma_i(Y_{\mathrm{hol}}).$$

If these bounds are uniform as a parameter tends to zero, any nonzero power-law orders of the holomorphic singular values are preserved.

Proof. For invertible A, B, the min–max characterization gives $\sigma_i(AYB) \le \|A\|_{\mathrm{op}}\|B\|_{\mathrm{op}}\sigma_i(Y)$. Apply this also to $Y = A^{-1}(AYB)B^{-1}$ to obtain the reverse bound. The operator norms of the inverse square roots are at most $m^{-1/2}$ and those of their inverses at most $M^{1/2}$; transpose has the same singular values. Multiplication by c yields the result. Uniform positive constants do not change a power-law order. □

Continuity and positive definiteness at a regular limiting point give such bounds locally. The preceding counterexample loses that lower bound as $\epsilon \to 0$. Conversely, an asymptotic order calculation alone does not fix the observed mass ratios or the point selected by stabilization. Published physical-flavour constructions are available [5]; that paper leaves stabilization and supersymmetry breaking untreated. This is a limitation of the result cited, not an assertion about every heterotic model in the literature.

6. **A quantitative simultaneous-persistence criterion**

The following theorem identifies bounds that would make a local construction robust. All its assumptions must be checked on the same compactification chart and in the same normalization.

THEOREM 6.1 (A certified interior minimum). *Let $B = \overline{B}_r(u_0) \subset \mathbb{R}^n$ and let V_0, Δ be C^2 on a neighborhood of B. Suppose on B that*

$$D^2V_0 \succeq \lambda I, \quad \|D^2\Delta\|_{\mathrm{op}} \le b < \lambda, \quad \|\nabla V_0(u_0)\| \le \eta, \quad \|\nabla\Delta(u_0)\| \le a.$$

Put $\kappa = \lambda - b$ and assume $\eta + a < \kappa r$. Then $V = V_0 + \Delta$ has exactly one critical point u_* in B. It is an interior minimum, $D^2V(u_*) \succeq \kappa I$, and

$$\|u_* - u_0\| \le \delta := \frac{\eta + a}{\kappa} < r. \tag{6.1}$$

Proof. The Hessian of V is bounded below by κI. Integrating along a segment yields strong monotonicity:

$$(\nabla V(x) - \nabla V(y)) \cdot (x - y) \ge \kappa \|x - y\|^2.$$

For x on the boundary and $v = x - u_0$ this gives $\nabla V(x) \cdot v \ge \kappa r^2 - (\eta + a)r > 0$. The inward directional derivative is negative, so a minimum on the compact ball cannot occur at its boundary. A minimum exists and is therefore critical in the interior. Strong monotonicity excludes two critical points. Apply it with $x = u_*$, $y = u_0$ and use $\nabla V(u_*) = 0$ to obtain $\kappa\|u_* - u_0\|^2 \le (\eta + a)\|u_* - u_0\|$, which proves (6.1). □

PROPOSITION 6.2 (Preserving masses and inequalities). *In Theorem 6.1, suppose Y is a physical 3×3 Yukawa matrix with $\|Y(u) - Y(u_0)\|_{\mathrm{op}} \le L\|u - u_0\|$ on B. If s_i are its singular values at u_0, then at u_* they differ from s_i by at most $e = L\delta$. In particular $s_{\min} > e$ certifies full rank. For $s_j > e$,*

$$\frac{\max(0, s_i - e)}{s_j + e} \le \frac{s_i(u_*)}{s_j(u_*)} \le \frac{s_i + e}{s_j - e}.$$

If a real acceptance function q is L_q-Lipschitz and has required lower bound q_{req}, then $q(u_0) > q_{\mathrm{req}} + L_q\delta$ certifies $q(u_*) > q_{\mathrm{req}}$.

Proof. The min–max formula for singular values and $|\|Av\| - \|Bv\|| \le \|(A - B)v\|$ on unit vectors give $|\sigma_i(A) - \sigma_i(B)| \le \|A - B\|_{\mathrm{op}}$ by applying the formula in both directions. The ratio bounds follow by bounding numerator and denominator separately. The last assertion is the Lipschitz inequality. □

Thus a controlled ball preserving the discrete spectrum and satisfying strict mass, hierarchy and consistency margins would turn an approximate minimum into a simultaneous certificate. If the real kinetic matrix obeys $0 < G(u_*) \preceq \Lambda I$, the physical scalar mass matrix $G(u_*)^{-1/2}D^2V(u_*)G(u_*)^{-1/2}$

has eigenvalues at least κ/Λ. This follows by applying the Hessian lower bound to $G^{-1/2}v$. At a critical point the connection correction to the Hessian vanishes. Thus kinetic normalization must also be bounded when interpreting the Hessian physically. The bounds must include all claimed perturbative and nonperturbative corrections, the normalization metrics, the physical moduli coordinates, and any constraints or gauge quotients. A Hessian estimate for a freely invented scalar function is not a Hessian estimate for the compactification potential.

7. **An explicit local effective-theory construction and its limit**

It is useful to separate a solvable analytic problem from the missing string realization. Consider a classical global $N = 1$ effective theory with fields $S, Z_1, \dots, Z_m$, constants $a > 0$, $F \neq 0$, and $m_j \neq 0$:

$$K = |S|^2 - a|S|^4 + \sum_j |Z_j|^2, \qquad W = FS + \frac{1}{2}\sum_j m_j Z_j^2.$$

We work on $|S|^2 < 1/(4a)$, where its Kähler metric is positive.

Proposition 7.1 (A stabilized local model). *For this specified classical global effective theory,*

$$V = \frac{|F|^2}{1 - 4a|S|^2} + \sum_j |m_j|^2 |Z_j|^2$$

has a unique minimum at the origin of the stated domain. Supersymmetry is broken there. In canonical real coordinates its Hessian has diagonal entries $4a|F|^2, 4a|F|^2$ *and* $|m_j|^2, |m_j|^2$ *for each* j.

Proof. $K_{S\bar{S}} = 1 - 4a|S|^2$ and $K_{Z_i\bar{Z}_j} = \delta_{ij}$, so the global-supersymmetry potential $K^{A\bar{B}} W_A \overline{W}_B$ is the displayed expression. Its first term is strictly increasing with $|S|^2$ and the remaining terms are positive quadratics. The auxiliary field in the S direction is nonzero at zero because $W_S = F$. Expanding with $S = (x + iy)/\sqrt{2}$ gives $V = |F|^2 + 2a|F|^2(x^2 + y^2) + \sum_j |m_j|^2(u_j^2 + v_j^2)/2 + O(|S|^4)$, which proves the Hessian assertion. □

This proposition supplies no Calabi–Yau bundle realizing its K and W. It omits supergravity, flux quantization, the anomaly constraints and the coupled visible sector. Consequently it verifies neither C7 nor C9 for the geometric example. Its role is to make the missing realization problem explicit rather than hide it inside a choice of coefficients.

PROPOSITION 7.2 (A tree-level obstruction). *Let a supergravity model have*

$$K = K_0(z,\bar{z}) - \sum_{i=1}^{r} n_i \log(T_i + \bar{T}_i), \quad n_i > 0, \quad \sum_i n_i = 3, \quad W = W(z).$$

At a point with $D_z W = 0$, its F-term potential vanishes for all T_i in this product chart. It does not stabilize these Kähler directions.

Proof. The T-block is diagonal, $K_i = -n_i/(T_i + \bar{T}_i)$ and $K_{i\bar{i}} = n_i/(T_i + \bar{T}_i)^2$. Hence $\sum_i K^{i\bar{i}} K_i K_{\bar{i}} = \sum_i n_i = 3$. Since $D_i W = K_i W$ and the z block contributes zero, these terms cancel the $-3|W|^2$ in $V_F = e^K(K^{A\bar{B}} D_A W \overline{D_B W} - 3|W|^2)$. Thus there are flat T directions. Additional D terms or corrections are not covered by this assertion and would need their own calculation. □

8. **The remaining proof obligations**

The geometric calculation supplies explicit nonstandard bundle data, cover-space polystability and anomaly effectiveness, and an exact three-family quotient index. It does not compute the full projected spectrum or all mass operators. In particular the quotient integral anomaly certificate and the hidden sector need a complete audit.

The local effective model proves that a specified mathematical potential can have a positive Hessian and broken supersymmetry. It does not show that this potential and the required physical Yukawa matrices arise from the same compactification. Theorem 6.1 provides a route for a controlled candidate: establish its hypotheses and the acceptance margins on that candidate's chart. Those estimates have not been supplied here.

Even an explicit metric-dependent selection function, a correct index, or a holomorphic hierarchy cannot replace these missing results. No intersection point of all nine viability loci has been established by the present constructions. The full GSC remains open in this draft.

Verification and provenance

Exact-arithmetic controls reproduce the Chern, slope and index data, the 256 invariant-system support cases, and the normalization and minimum examples. Sympy checks the local Hessian and no-scale identity. These are algebra checks, not a nine-condition construction; no Lean formalization is claimed.

ChatGPT/Codex assisted with this draft at the author's request. Both existing authors are retained, and their review of the new material is pending. No v2 has been uploaded to Hexagon.

References

[1] D. Bhattacharjee, U. Bhattacharya and S. Ghosh, *One missing axiom for the complete closure of the heterotic landscape*, Preprints.org (2026), version 2, doi:10.20944/preprints202603.0792.v2. Cited for the formulation of the nine conditions, not as a proved existence theorem.

[2] D. Bhattacharjee and U. Bhattacharya, *Which string vibrations become which generations on the Dwork quintic*, Hexagon 2610.00446v1; code archive creelie/strings v1.2.0.

[3] D. Bhattacharjee and U. Bhattacharya, *The mirror clause of the Geometric Selection Conjecture is empty*, Hexagon 2610.00493v1; doi:10.5281/zenodo.23567115.

[4] E. I. Buchbinder, A. Constantin and A. Lukas, *The moduli space of heterotic line bundle models: a case study for the tetra-quadric*, J. High Energy Phys. **03** (2014), 025, doi:10.1007/JHEP03(2014)025. The bundle in (4.1) is the published example of Section 5.1.

[5] A. Constantin, L. T.-Y. Leung, A. Lukas and L. A. Nutricati, *Reproducing Standard Model fermion masses and mixing in string theory: a heterotic line bundle study*, Phys. Rev. D **113** (2026), 046005, doi:10.1103/fbdz-t73z.

[6] R. Hartshorne, *Algebraic Geometry*, Graduate Texts in Mathematics 52, Springer, 1977.

[7] J. M. Lee, *Introduction to Smooth Manifolds*, second ed., Graduate Texts in Mathematics 218, Springer, 2013.

Deep Bhattacharjee
Formerly, Electro-Gravitational Space Propulsion Laboratory, Bhubaneswar, India; formerly, University of Calcutta, Kolkata, India
E-mail: itsdeep@live.com

Ushashi Bhattacharya
Formerly, National Taiwan University, Taipei, Taiwan
E-mail: rai.bhattacharya.in@gmail.com